UNIT V — Operations Management and Supply Chain

1. Introduction to Operations Management

Operations Management (OM) is the set of activities, decisions, and responsibilities that convert inputs (materials, labor, capital, information) into goods and services that create value for customers. OM sits at the heart of any organization that produces goods or delivers services: it determines how resources are organized, how work flows, and how performance is measured.

Why it matters: OM governs cost, quality, speed, flexibility, and reliability — the operational levers that determine whether a company can deliver promised value profitably and repeatedly. Poor operations increase costs, cause delivery failures, and erode customer trust; excellent operations create competitive advantage through lower costs, faster time-to-market, and superior reliability.

Core functions and activities

- **Process design:** Defining how work is done (sequence of activities, layout, handoffs).
- **Capacity planning:** Determining production/service capacity to meet demand (short, medium, long term).
- **Facilities & layout:** Physical arrangement of equipment, workstations, and flow paths to minimize waste and delay.
- **Workforce planning & scheduling:** Hiring, training, rostering, and managing labor to meet service/product demand.
- **Supply management & procurement:** Ensuring inputs are available, cost-effective, and of required quality.
- **Quality assurance & improvement:** Defining quality standards, monitoring, and continuous improvement.
- **Performance measurement & control:** Setting KPIs and feedback loops (productivity, utilization, cycle time).

Process types

- **Project processes** (unique, temporary — e.g., construction)
- **Job shop / intermittent** (customized low-volume)
- **Batch process** (medium volume, variety)
- **Assembly line / flow** (high volume, standardized)
- **Continuous process** (very high volume, 24/7 operations)

Key performance metrics

- **Throughput:** Units produced per time period.
- **Cycle time:** Time to complete one unit or transaction.
- **Utilization:** % of capacity actually used.
- **Productivity:** Output per input (labor hours, machine hours).
- **On-time delivery:** % delivered on schedule.
- **First-pass yield / defect rate.**

Analytical tools & techniques

- **Process mapping and value-stream mapping (VSM)** to identify value-added vs non-value-added steps.
- **Bottleneck analysis** (identify constraints and relieve them).
- **Little's Law:** W=L/λW = L/\lambdaW=L/λ (or commonly Inventory=Throughput×Lead TimeInventory = Throughput \times Lead\;TimeInventory=Throughput×LeadTime), used to relate work-in-process, throughput, and cycle time.
- **Simulation, queuing models, and capacity modelling** for service operations.

Common pitfalls

- Designing processes without validating customer needs.
- Over-capacity or chronic under-capacity due to poor forecasting.
- Focusing on utilization at the expense of flow (high utilization can increase queues and cycle time).
- Ignoring variation; variability is the major cause of delays and poor service.

Practical advice

- Define customer "order winners" (what wins business) and "order qualifiers" (minimum acceptable criteria) and design operations around them.
- Start with process mapping to visualize waste.
- Balance capacity, not to maximize utilization but to optimize flow and responsiveness.

2. Supply Chain Management

Supply Chain Management (SCM) is the coordinated management of all activities involved in sourcing, procurement, conversion, and logistics to deliver products from suppliers to end customers. SCM spans suppliers, manufacturers, distributors, retailers, and customers — and integrates material, information, and financial flows.

Why it matters: SCM links strategic sourcing to customer satisfaction and profitability. Efficient supply chains lower costs, improve service levels, reduce inventory, and provide resilience against disruptions.

Core components

- **Sourcing & procurement:** Supplier selection, contracting, negotiation, supplier performance management.
- **Manufacturing / conversion:** Turning inputs into finished goods (in-house or outsourced).
- **Logistics & distribution:** Transportation, warehousing, order fulfillment, last-mile delivery.
- **Demand planning & forecasting:** Predicting customer demand to drive supply decisions.
- **Inventory management:** Balancing service levels with carrying costs.
- **Returns management / reverse logistics:** Handling returns, repairs, and recycling.
- **Information flows & IT systems:** ERP, TMS, WMS, EDI, and modern APIs enabling visibility.

Frameworks

- **SCOR model (Supply Chain Operations Reference):** Plan → Source → Make → Deliver → Return → Enable; used to standardize processes and metrics.
- **Push vs Pull strategies:** Push (forecast-driven) vs pull (demand-driven/JIT); many modern supply chains use hybrid approaches.

Common phenomena & how to mitigate

- **Bullwhip effect:** Small demand changes at retail amplify upstream. Mitigation: share POS data, reduce lead times, implement vendor-managed inventory (VMI), coordinate planning (CPFR — collaborative planning, forecasting, and replenishment).
- **Single-source risk:** Diversify suppliers, qualify backups, and maintain safety stock strategically.
- **Lead time variability:** Shorten lead times through local sourcing, supplier development, or faster logistics.

Technology & analytics

- **S&OP / IBP (Integrated Business Planning):** Monthly cross-functional planning aligning demand, supply, and finance.
- **Advanced planning systems (APS):** Optimize production and distribution planning.
- **Visibility tools:** Real-time tracking via IoT, RFID, and cloud-based platforms.
- **Analytics:** Forecasting models, scenario planning, inventory optimization.

Sustainability & ethics

- Increasing emphasis on green supply chains, traceability (conflict minerals, labor practices), carbon footprint reporting, and circular economy practices.

Metrics commonly tracked

- **Fill rate, perfect order rate, on-time delivery, inventory turns, days of inventory, total landed cost.**

Practical checklist

- Map end-to-end supply chain flows and measure lead time at each step.
- Implement basic S&OP and focus on improving forecast accuracy.
- Negotiate SLAs with key suppliers and track supplier KPIs.
- Build contingency plans for top suppliers and logistics lanes.

3. Inventory Management

Inventory management is the set of policies and processes that control the acquisition, storage, and movement of inventory to meet customer demand at acceptable cost and service levels. Inventory ties up working capital, so managing it well improves cash flow and responsiveness.

Types of inventory

- **Raw materials** (inputs to production)
- **Work-in-process (WIP)** (in-production items)
- **Finished goods** (ready for sale)
- **Transit inventory** (in the logistics pipeline)
- **Safety stock** (buffer against variability)
- **Decoupling stock** (buffers between operations)

Cost trade-offs

- **Holding (carrying) cost:** Storage, insurance, obsolescence.
- **Ordering cost:** Administrative and setup costs per order.
- **Stockout cost:** Lost sales, backorder costs, customer dissatisfaction.

Inventory control policies

- **Continuous review (Q system):** Monitor inventory continuously and order EOQ when ROP is reached.
- **Periodic review (P system):** Review inventory at fixed intervals and order up to a target level.
- **ABC analysis:** Classify SKUs by value/usage: A (high value, tight control), B (moderate), C (low value, loose control).
- **Service-level driven policies:** Set safety stock to achieve desired fill rates or cycle-service levels.

Modern approaches

- **Just-In-Time (JIT):** Minimize inventory by synchronizing supplies with production (requires reliable suppliers and short lead times).
- **Vendor-Managed Inventory (VMI):** Supplier monitors and replenishes inventory at buyer sites.
- **Cross-docking:** Minimize warehousing by transferring inbound goods directly to outbound vehicles.

Key KPIs

- Inventory turnover = Cost of Goods Sold / Average Inventory.
- Days of Inventory (DOI) = 365 / Inventory turnover.
- Fill rate, stockout frequency, carrying cost as % of inventory value.

Common mistakes

- Over-investing in safety stock without addressing root causes of variability.
- Using same policy for all SKUs (ignore ABC differentiation).
- Poor data quality (inaccurate demand history) leads to wrong safety stocks and ROPs.

Practical tips

- Clean and reconcile inventory records regularly (cycle counting).
- Use ABC classification to prioritize improvement efforts.
- Improve forecast accuracy through better demand signals (POS, market intelligence).
- Run experiments when changing reorder policies and measure impact on fill rates and carrying costs.

4. Lean Manufacturing

Lean manufacturing is a philosophy and a set of practices aimed at maximizing customer value while minimizing waste. Originating from the Toyota Production System (TPS), Lean focuses on improving flow, reducing lead times, and continuously eliminating non-value-adding activities.

Core principles (Womack & Jones)

1. **Specify value** from the customer perspective.
2. **Map the value stream** and identify waste.
3. **Create continuous flow** so work moves without delay.
4. **Establish pull** so production reacts to customer demand.
5. **Pursue perfection** through continuous improvement.

Types of waste (TIMWOODS / 8 wastes)

- Transportation, Inventory, Motion, Waiting, Overproduction, Overprocessing, Defects, Skills (underutilized talent).

Key tools & methods

- **5S (Sort, Set in order, Shine, Standardize, Sustain):** Workplace organization.
- **Value Stream Mapping (VSM):** Visualizes the flow of materials and information to identify bottlenecks and waste.
- **Kanban:** Visual pull system for inventory replenishment and shop-floor control.
- **Kaizen:** Continuous improvement workshops focused on incremental changes.
- **SMED (Single-Minute Exchange of Dies):** Reduce setup/changeover times to enable smaller lot sizes.
- **Poka-Yoke:** Mistake-proofing devices to prevent defects.
- **TPM (Total Productive Maintenance):** Maximize equipment effectiveness via preventive maintenance and operator involvement.
- **Heijunka:** Level production to reduce variability and smooth demand.

Operational impacts

- Reduced lead times, smaller batch sizes, lower inventory, improved quality, faster takt time alignment, and improved responsiveness.

Metrics

- **Overall Equipment Effectiveness (OEE):** Availability × Performance × Quality.
- **Lead time, throughput, defect rates, changeover times.**

Implementation approach

1. Start with leadership commitment and training.
2. Map current state (VSM), identify quick wins, and pilot on a line.
3. Establish standard work and metrics.
4. Spread practices across the plant and embed Kaizen routines.
5. Sustain via training, audits, and recognition.

Pitfalls

- Treating Lean as a set of tools rather than a cultural transformation.
- Forcing flow when process variability or supplier unreliability exist — fix variability first.
- Ignoring human factors; Lean requires engagement and ownership from frontline staff.

Practical guidance

- Run rapid Kaizen events to solve visible problems and build momentum.
- Reduce changeover times aggressively to enable smaller batches and reduce WIP.

- Use Kanban to synchronize downstream demand with upstream supply.

5. Operations Planning and Scheduling

Operations planning and scheduling turn strategic objectives into executable plans — deciding what to produce, when, and with what resources. Sound planning ensures capacity and materials match demand while scheduling allocates resources at the operational level.

Planning hierarchy

- **Strategic planning:** Long-term capacity decisions, facility location, capital investments.
- **Tactical planning / Aggregate planning:** Medium-term balancing of capacity and demand (3–12 months) — decisions on workforce levels, inventory buffers, subcontracting.
- **Operational planning / Master Production Schedule (MPS):** Short-term schedule for finished goods.
- **Detailed scheduling / shop-floor scheduling:** Assign specific jobs to machines/people hourly/daily.

Techniques & systems

- **MRP (Material Requirements Planning):** Time-phased requirement planning that explodes MPS into component requirements using BOMs and lead times. Inputs: MPS, BOM, inventory records, lead times. Outputs: planned orders, order releases.
- **ERP systems:** Integrate financials, procurement, production, and inventory into a single database to enable planning and execution.
- **APS (Advanced Planning & Scheduling):** Finite capacity planning and optimization for complex manufacturing.
- **S&OP / IBP:** Cross-functional process aligning demand and supply, reconciling financial plans with operational constraints.

Scheduling rules & heuristics

- **FCFS (First Come First Served), SPT (Shortest Processing Time), EDD (Earliest Due Date)** — used for prioritized sequencing.
- **Johnson's rule:** Optimal sequencing for two-machine flow-shop problems to minimize makespan.
- **Drum-Buffer-Rope (TOC):** Protect the constraint (drum) with buffers to synchronize flow and use rope to pace upstream release.
- **Finite vs Infinite scheduling:** Finite scheduling respects resource availability; infinite ignores it.

Key metrics

- Makespan, throughput, on-time delivery, lead time adherence, machine utilization, backlog.

Practical planning advice

- Use aggregate planning to smooth peaks (overtime, subcontracting, seasonal inventory).
- Keep BOMs and lead times updated — MRP is only as good as its data.
- Apply finite scheduling in bottleneck-prone environments to avoid promising impossible lead times.
- Maintain a short-term control loop for schedule changes and expedite/containment policies.

Common issues

- Poor data quality (inaccurate inventory or lead times) leading to late orders or excess stock.
- Over-reliance on MRP without capacity checks — creates “never-ending” expediting.
- Lack of coordination between sales and operations — misaligned promises.

6. Quality Management and Continuous Improvement

Quality management ensures products and services meet customer requirements and regulatory standards. Continuous improvement is the ongoing effort to raise performance, remove defects, and increase value.

Foundational approaches

- **Total Quality Management (TQM):** Organization-wide commitment to quality, customer focus, and continuous improvement.
- **Six Sigma:** Data-driven method to reduce variation and defects; DMAIC (Define, Measure, Analyze, Improve, Control) is the core cycle.
- **PDCA (Plan-Do-Check-Act):** Iterative improvement cycle.
- **ISO 9001:** International standard for quality management systems — emphasizes process approach, documentation, and continual improvement.

Analytical tools

- **Statistical Process Control (SPC):** Uses control charts ($\bar{X}$, R, p-charts) to monitor process stability and detect special-cause variation.
- **Process capability indices:** CpC_pCp and CpkC_{pk}Cpk measure how well a process fits within specification limits.
- **Root cause analysis:** 5 Whys, Ishikawa (fishbone) diagram, fault-tree analysis.
- **Design of Experiments (DOE):** Systematic testing to discover factor effects and optimal settings.

Cost of quality

- **Prevention costs:** Training, process control, design for quality.
- **Appraisal costs:** Inspection, testing.
- **Internal failure costs:** Scrap, rework.
- **External failure costs:** Warranty claims, returns, lost customers.

Key metrics

- Defects per million opportunities (DPMO), yield, PPM (parts per million), cost of poor quality, customer satisfaction (CSAT), Net Promoter Score (NPS).

Continuous improvement culture

- Encourage frontline ownership of problems, standardize successful improvements, and institutionalize Kaizen and A3 problem solving. Use visual management and daily performance huddles to surface issues.

Practical steps to implement

1. Establish quality objectives aligned with business goals.
2. Map key processes and define critical-to-quality (CTQ) characteristics.
3. Train teams in basic quality tools and problem-solving.
4. Use SPC to monitor processes and trigger countermeasures when charts indicate out-of-control signals.
5. Run DMAIC projects for complex problems and sustain gains with control plans.

Common pitfalls

- Confusing inspection with quality improvement — inspection finds defects but does not prevent them.
- Launching many improvement projects without ensuring follow-through and controls.
- Not involving operators in problem-solving — loses valuable insight and buy-in.

7. Logistics and Distribution Management

Logistics and distribution manage the physical movement and storage of goods from origin to final customer. It covers transportation, warehousing, order fulfillment, and the systems that coordinate these activities.

Core functions

- **Transportation management:** Selecting modes (air, sea, rail, road), carriers, routing, and consolidation strategies.
- **Warehousing:** Receipt, storage, picking, packing, and shipping. Warehouse design affects throughput and accuracy.

- **Order fulfillment:** Picking methods (zone, wave, batch), packing, labeling, and speed to customer.
- **Network design:** Location and number of distribution centers to balance cost and service.
- **Reverse logistics:** Handling returns, refurbishing, recycling and disposal.

Transportation considerations

- **Trade-offs:** Speed vs. cost; air is fastest/most costly, sea is cheap/slower, intermodal combines.
- **Consolidation & LTL vs FTL:** Consolidate small shipments to reduce cost; LTL (less-than-truckload) vs FTL (full truckload) decisions.
- **Last-mile delivery:** Most expensive leg; optimization focuses on density, micro-hubs, dynamic routing, locker networks, and partnering with local couriers.

Warehousing strategies

- **Cross-docking:** Minimal warehousing — inbound sorted for immediate outbound shipping.
- **Dedicated vs shared storage:** Dedicated space for high-turn SKUs; shared for flexibility.
- **Automation:** Conveyor systems, AS/RS (automated storage and retrieval systems), robotics for high-volume operations.
- **Pick strategies:** Choose methods to optimize accuracy and speed: batch, zone, wave, or pick-to-light.

Technology

- **WMS (Warehouse Management System)** controls inventory location, putaway, picking and replenishment.
- **TMS (Transportation Management System)** plans and optimizes routing, tendering, and freight accounting.
- **Route optimization and telematics** improve utilization and reduce fuel/CO2.

Compliance & trade

- Customs clearance, incoterms, tariff classification, and documentation management are critical for international distribution.

Risk & resilience

- Build redundancy in carriers and routes, plan for seasonal surges, and maintain contingency stocks. Use dynamic rerouting and real-time tracking to handle disruptions.

KPIs

- On-time delivery, cost per shipped unit, order cycle time, perfect order rate, inventory dwell time, freight cost as % of sales.

Sustainability trends

- Consolidation, modal shift to lower-carbon modes, eco-packaging, and carbon accounting for logistics operations.

Practical checklist

- Design the network to meet target transit times at acceptable cost.
- Use TMS/WMS to automate routine processes and increase visibility.
- Optimize last-mile using density and micro-hubs, and measure the cost-to-serve per zone.
- Plan for reverse logistics and have clear returns policies and procedures.